

11 (**short**) An electron microscope is situated in the basement of a building. It is sensitive to horizontal vibration and so is vibration isolated as shown in Fig. 8. The microscope is modelled as a rigid mass m whose displacement is y . The basement motion is x . The total isolation stiffness and damping are $2k$ and 2λ respectively.

(a) Find a differential equation for the motion y of the microscope given the basement motion x . [4]

(b) The basement is moving sinusoidally according to $x(t) = X \cos \omega t$ and motion of the microscope likewise is $y(t) = Y \cos \omega t$. Use the Mechanics Data Book or otherwise to sketch a graph of the ratio Y/X versus frequency ω . [6]

For part (b) use $m = 1000 \text{ kg}$, $k = 8 \text{ N / m}$ and $\lambda = 400 \text{ N s / m}$.

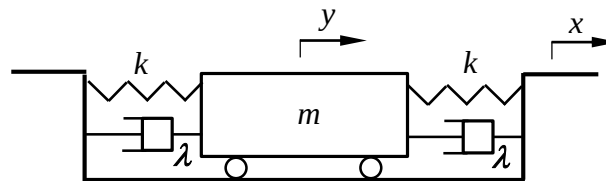


Fig. 8